

Adaptive Intelligent Vehicle Safety System Using Ultrasonic-Based Obstacle Detection and LED Zone Control

Eesha.D

Department of Electronics and
Communication
B M S College of Engineering
Bengaluru, India
eesha.ece@bmsce.ac.in

Kovuru Sai Karthik

Department of Electronics and
Communication
B M S College of Engineering
Bengaluru, India
kovurusai.ec23@bmsce.ac.in

Manish Kumar Bale

Department of Electronics and
Communication
B M S College of Engineering
Bengaluru, India
manishkumarbale.ec23@bmsce.ac.in

Madhusudhan C

Department of Electronics and
Communication
B M S College of Engineering
Bengaluru, India
madhusudhanc.ec23@bmsce.ac.in

Abstract—This project presents an Intelligent Vehicle Safety System that enhances night-time driving safety through adaptive illumination and real-time obstacle detection. Dual ultrasonic sensors mounted on independent servo mechanisms perform directional scanning to identify obstacle distance and angular position. Based on this information, the microcontroller dynamically adjusts WS2812 LED headlight intensity through zone-based dimming to minimize glare while maintaining visibility. A buzzer alert warns the driver when obstacles fall below critical distances. Testing demonstrated stable scanning, obstacle detection accuracy of 3–5 cm, and LED response times under 70 ms. The system offers a low-cost, scalable embedded approach comparable to essential ADAS lighting functionalities.

Keywords—Adaptive headlights, ultrasonic sensors, servo scanning, LED zone control, vehicle safety.

I. INTRODUCTION

Night-time driving presents a substantially elevated risk compared to daytime operation, largely attributable to reduced visibility and the limitations of conventional vehicle lighting systems. Statistics consistently show that a disproportionately high number of fatal accidents occur during low-light hours, emphasizing the critical need for enhanced vehicular safety technologies [1]. Over the last decade, the automotive industry has responded with sophisticated Advanced Driver Assistance Systems (ADAS), which use integrated sensors and processing units to improve driver awareness, enable collision avoidance, and, critically, manage vehicle illumination. Adaptive lighting is a cornerstone of ADAS, aiming to actively shape the light beam in response to dynamic driving conditions.

The primary limitation of traditional fixed-beam headlights is their inability to adapt to the driving environment. These systems provide uniform, static illumination regardless of road curvature, vehicle speed, or the presence of other road users. This inflexibility leads to two major safety drawbacks. First, the fixed pattern often results in insufficient object visibility in off-center areas, particularly around bends or in critical blind spots. Second, fixed high-beam headlights are the leading cause of temporary driver glare and visual

impairment for oncoming motorists, necessitating frequent and often delayed manual adjustments by the driver. The failure of conventional systems to provide adequate safety and visibility under complex, dynamic conditions establish the fundamental problem addressed by modern adaptive lighting.

Contemporary commercial Adaptive Driving Beam (ADB) systems utilize complex and expensive sensory input, primarily relying on cameras, LiDAR (Light Detection and Ranging), and radar units [3], [9]. Camera-based solutions perform detailed image processing to detect the exact position of oncoming headlights and preceding taillights, allowing for the creation of precise, non-glare "shadows" within the light beam [2]. While highly effective, the required sensor suite, high-speed processing unit (GPU/FPGA), and complex calibration procedures translate directly into high material and manufacturing costs. Consequently, these sophisticated ADAS illumination technologies remain predominantly restricted to high-end and luxury vehicles, excluding a vast majority of the global fleet—particularly low- and mid-range automobiles—from these vital safety improvements. This economic accessibility gap in adaptive lighting technology represents a major challenge in democratizing automotive safety.

This project introduces an Adaptive Intelligent Vehicle Safety System that bridges this gap by demonstrating that essential adaptive functionality can be achieved using low-cost, embedded hardware and microcontroller-based logic. The core of the proposed system is a novel approach to directional object detection. Instead of expensive optical or radar-based sensors, the system employs dual ultrasonic sensors mounted on independent servo mechanisms. This servo-assisted angular scanning allows the system to overcome the limitation of fixed-position ultrasonics by providing real-time data on both the distance and the relative angular position of obstacles—a crucial requirement for directional illumination. This low-cost sensing technique successfully replicates the *angular awareness* provided by far more expensive ADAS components.

The second key innovation lies in the adaptive illumination mechanism. The headlight is implemented using programmable WS2812 LED strips, which are subdivided into distinct, controllable illumination zones. Based on the output from the directional

ultrasonic scan, the microcontroller (an Arduino platform) dynamically adjusts the intensity of specific LED zones. For example, if an obstacle or oncoming vehicle is detected in the 30° left zone, the corresponding LED zone is dimmed to prevent glare, while the remaining zones maintain high illumination for maximum road visibility. A buzzer module is integrated to provide a secondary, immediate auditory warning to the driver when an obstacle enters a pre-defined critical distance zone (e.g., less than 60 cm), enhancing the overall safety feedback loop. This system architecture demonstrates how microcontroller-based control and low-latency feedback can replicate the real-time adjustments characteristic of high-end ADAS.

The primary objective of this work is to design, implement, and validate a highly accessible intelligent vehicle safety system that provides adaptive illumination and obstacle detection. The system aims to achieve three specific goals:

1. To successfully integrate servo-assisted ultrasonic scanning for reliable, two-dimensional mapping of the immediate forward environment.
2. To implement a zonal LED dimming strategy that dynamically adjusts headlight output based on the detected angular position of obstacles.
3. To confirm system performance metrics, including detection accuracy (targeting 3–5 cm error) and low response latency (under 70 ms), suitable for real-time safety applications. The successful demonstration of this system validates a path toward integrating essential ADAS features into entry-level vehicles, significantly enhancing night-time safety without the prohibitive cost burden.

The remainder of this paper is structured as follows. Section II presents a Literature Survey detailing existing work in adaptive lighting, sensor fusion, and low-cost embedded solutions. Section III defines the Problem Analysis and the specific design challenges addressed. Section IV outlines the Methodology and Implementation, including the system architecture, hardware components, and firmware logic. Section V presents the Results and Discussion, summarizing the performance tests on obstacle detection, accuracy, and LED response time. Finally, Section VI provides the Conclusion and Future Trends of this work.

II. LITERATURE SURVEY

Adaptive lighting technology has undergone substantial evolution over the last decade, reflecting advancements in sensing, embedded control, and automotive safety systems. Comprehensive surveys in the field emphasize that modern driving safety relies heavily on the convergence of sensing, communication, and intelligence to mitigate risks such as collisions and poor visibility [1].

Early vehicle headlight systems relied purely on mechanical mechanisms and fixed illumination patterns, offering no adaptability to road curvature, obstacle proximity, or oncoming traffic. Research progressed toward automatic headlight dimming systems based on photodetectors; however, these systems lacked directional understanding and operated solely on the intensity of incoming light. They were unable to distinguish between actual obstacles, ambient reflections, or road geometry. Such systems, while simple and cost-effective, did not provide significant safety improvements in complex night-time driving conditions [15].

Modern adaptive headlight systems incorporate sophisticated sensors such as cameras, infrared sensors, radar, and LiDAR. For instance, recent studies have proposed intelligent headlight systems utilizing Digital Micromirror Devices (DMD) and sensor fusion to achieve precise beam shaping [2]. Camera-based systems enable lane detection, pedestrian identification, and vehicle recognition through image processing algorithms [7], [12]. Furthermore, fusing data from radar and cameras has become a standard approach for robust object detection and distance estimation in Advanced Driver Assistance Systems (ADAS) [3], [10]. These technologies allow vehicles to autonomously identify approaching vehicles, evaluate relative speed, and adjust both horizontal and vertical beam distribution in real time [6].

The integration of these sensors is not limited to lighting alone; it often forms part of a holistic safety architecture. Research demonstrates that data used for adaptive front-lighting systems (AFS) is frequently shared with stability control [4] and pre-crash braking assistants [11] to improve overall vehicle dynamics and reaction times. Similarly, multi-sensor fusion frameworks have been developed to coordinate collision avoidance maneuvers with adaptive lighting adjustments [5], [8], [9]. However, integrating such systems requires complex calibration, high computational capacity, and expensive hardware components, making them unsuitable for economical or student-level implementations.

To address cost barriers, several studies have examined the use of simpler embedded solutions. Research into smart electric mobility systems has demonstrated the viability of integrated safety and headlight monitoring using accessible components [13]. Ultrasonic technology, extensively used in parking assistance, performs reliably in short-range environments, even in low-light or night conditions [14]. However, most literature focuses on fixed-position ultrasonic modules or basic photodetector setups, which limits their ability to provide the angular information necessary for true adaptive lighting.

Servo-assisted angular scanning has emerged as a promising method for expanding the spatial awareness of ultrasonic sensors. By rotating the sensor across a predefined angular sweep, a two-dimensional perception of the environment can be constructed. While research in robotics has utilized this for mapping, its direct integration with automotive adaptive lighting remains limited.

Another stream of literature focuses on LED-based adaptive lighting. With advancements in LED efficiency, researchers have explored adaptive illumination using LED matrices. Studies highlight the importance of zonal dimming, where different regions of the headlight beam can be independently controlled to avoid dazzling other drivers while maintaining illumination elsewhere [2], [6]. This mirrors the behavior of high-end matrix LED headlights but is often implemented in academic prototypes using low-cost microcontrollers. However, many of these prototypes lack the robust environmental sensing found in commercial ADAS, relying instead on simple LDRs or static sensors.

In summary, the literature reveals extensive development in adaptive lighting through high-end technologies like sensor fusion [3], [9] and modest development in low-cost embedded alternatives. The clear research gap lies in creating an affordable, yet functionally effective system that integrates wide-field obstacle detection with precise illumination control. This project directly addresses that gap.

III. PROBLEM ANALYSIS AND SOLUTION

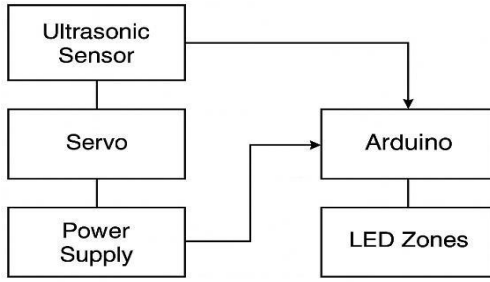


Fig. 1. Hardware architecture

Nighttime driving is one of the leading contributors to road accidents worldwide due to poor visibility, unpredictable obstacles, and inconsistent driver reaction times. Traditional headlights, designed with a fixed beam distribution, fail to adapt to changing driving environments. They illuminate the road uniformly regardless of the location of obstacles, pedestrians, or reflective surfaces. As a result, drivers may experience severe glare when light reflects off oncoming vehicles, wet surfaces, or nearby objects. The lack of dynamic illumination negatively impacts visibility and increases the cognitive load on the driver, leading to slower reaction times and higher accident risk.

Although modern vehicles incorporate Adaptive Driving Beam (ADB) systems and intelligent headlight controls, these rely on high-cost sensors such as cameras, radar units, and LiDAR. These systems require sophisticated image processing algorithms, high-speed computational hardware, and precision optical assemblies. Their dependency on multiple high-cost modules makes them inaccessible for budget-constrained vehicles, student projects, and lower-end automotive markets. Additionally, their sensitivity to adverse weather conditions such as fog, rain, or dust reduces overall reliability.

Existing low-cost solutions—such as LDR-based dimming circuits—are fundamentally limited. They react only to incoming light intensity and cannot identify object locations, distances, or angular positions. Ultrasonic distance sensors, while widely used, traditionally measure only along a fixed axis. Without angular scanning, they cannot provide the directional information necessary for dynamic beam shaping.

Thus, the fundamental problem is the absence of a low-cost, wide-field sensing and adaptive illumination system capable of detecting obstacles, determining their angular position, and adjusting lighting output in real time.

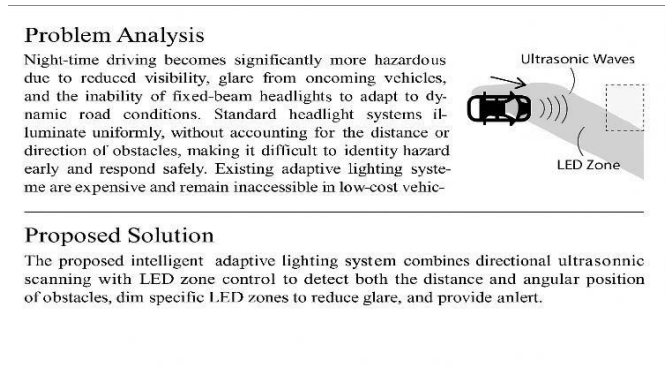


Fig. 2. Proposed Solution and Analysis

The proposed system solves this problem by combining servo-assisted ultrasonic scanning with programmable LED zonal illumination. Instead of relying on fixed sensors, ultrasonic

sensors are mounted on servo motors that continuously sweep across a 180° field of view. This scanning method generates distance measurements at multiple angles, providing a panoramic understanding of the environment. The system then maps these angles to corresponding LED zones, dimming specific regions of the headlight beam whenever obstacles are detected. This dynamic illumination replicates the behavior of modern matrix LED headlights while using extremely low-cost components.

The LED dimming system ensures that illumination remains strong in obstacle-free regions, improving overall visibility. When objects are close, the system activates a buzzer to alert the driver immediately. This multi-layered safety approach, combining visual adaptation, spatial awareness, and audible alerts, makes the solution effective for night driving at a fraction of the cost of ADB systems.

IV. METHODOLOGY AND IMPLEMENTATION

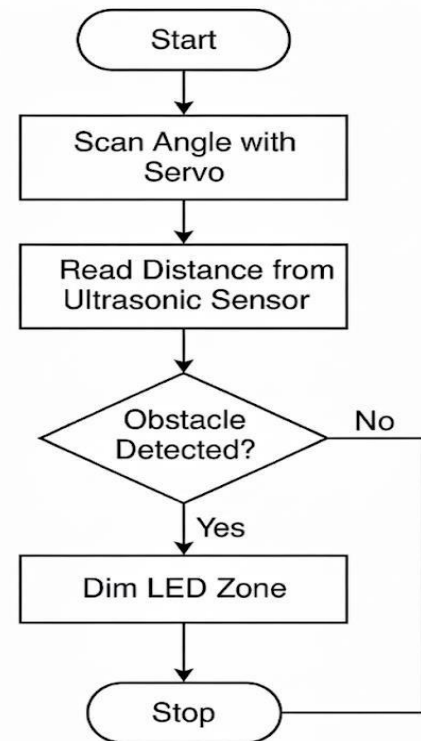


Fig. 3. Operational Flowchart

The methodology adopted in this project focuses on creating a unified embedded system capable of detecting obstacles, determining their angular position, and adjusting illumination in real time to enhance night-time driving safety. The operation begins with the integration of two ultrasonic sensors mounted on independent servo motors. These servo motors continuously sweep from 0° to 180°, pausing momentarily at each angle to capture distance readings. This approach allows the system to scan a wide field of view instead of relying on a single forward-facing measurement, enabling the identification of obstacles appearing from different directions. The angular and distance data collected during this sweep forms the basis for understanding the obstacle's position relative to the vehicle's line of motion. The captured readings are then processed by the Arduino controller, where filtering and validation help eliminate noise caused by reflective surfaces, sensor jitter, or environmental disturbances.

Once valid sensor data is obtained, the system maps each angle to a corresponding illumination zone on the WS2812 LED strip. The headlight is divided into multiple zones, each representing a region within the servo's scanning range. This zonal structure enables the system to adjust brightness only in the areas where obstacles are

detected instead of altering the entire headlight beam. When an obstacle appears within a caution or danger distance, the LED brightness of its corresponding zone is reduced to minimize glare while maintaining adequate illumination in other directions. This selective dimming mechanism provides a real-time adaptive lighting effect similar to that found in high-end automotive systems but implemented using low-cost hardware. The LED adjustments are executed through programmable control using the NeoPixel library, allowing fast and smooth visual transitions.

A summary of the key operational parameters is presented in Table I-System operational parameters offering a clear overview of system configuration and threshold values used during implementation.

Parameter	Value
Servo Sweep Range	0° – 180°
Step Resolution	~1° per step
Ultrasonic Detection Range	2 – 400 cm
Caution Distance Threshold	120 cm
Danger Distance Threshold	60 cm
LED Zones	6–8 zones depending on mapping
Loop Cycle Time	~100–150 ms

TABLE I

In addition to illumination control, the system incorporates an audio alert using a simple buzzer. When the sensed distance falls below a predefined danger threshold, the buzzer activates to warn the driver of an immediate hazard. This enhancement ensures that even if the driver fails to notice the visual dimming, an audible alert will prompt immediate attention. Power distribution is managed through a buck converter that steps down the 12-volt supply to a stable 5-volt output required by the ultrasonic sensors, servo motors, microcontroller, and LEDs. Proper grounding and current management ensure reliable operation of all components without overheating or voltage fluctuations. The entire system runs on a continuous control loop that coordinates sensor scanning, data processing, illumination adjustments, and alert triggering, enabling seamless real-time behavior during operation.

Beyond the core sensing and illumination mechanisms, careful consideration was given to the physical arrangement and integration of the hardware components. The ultrasonic sensors were mounted on servo brackets to ensure stable and repeatable rotation during scanning. Each servo was calibrated to maintain consistent angular steps, preventing drift or misalignment during extended operation. The WS2812 LED strip was positioned to emulate a multi-zone headlight structure, with zones mapped proportionally to the servo sweep range. Wiring between the sensors, servos, LED strip, and Arduino was organized to minimize electrical

noise and ensure reliable signal transmission, while the buck converter provided stable power regulation to prevent voltage drops during simultaneous servo and LED activity.

On the software side, implementation relied on a structured control loop designed to operate continuously with minimal latency. The loop begins with initiating the servo sweep, followed by acquiring ultrasonic readings at each angular step. Each reading undergoes validation using basic noise-suppression techniques such as median filtering and rejection of outlier values produced by irregular reflections. This ensures that transient spikes or false echoes do not influence the dimming or alert logic. The servo delay between angle increments was optimized to balance scanning resolution and system responsiveness, resulting in sufficient environmental coverage without introducing noticeable lag.

Once filtered distance values are obtained, the decision-making module maps each angle to its corresponding LED zone and evaluates whether the measured distance falls into the caution or danger threshold. LED brightness updates are executed immediately using the NeoPixel library, which offers precise control of individual LEDs even under rapid consecutive updates. To enhance safety, the buzzer logic is integrated within the same loop, activating whenever the sensed distance falls below the critical limit. This ensures that all alert mechanisms remain synchronized with the sensing cycle.

The final integrated system performs consistently with minimal latency, allowing rapid adjustments to environmental changes. By combining synchronized scanning, validated data acquisition, and zonal LED response, the methodology ensures that the system behaves predictably under varying conditions. The modular nature of the design also allows easy refinement, such as increasing zone resolution, integrating additional sensors, or extending the scanning field, making the implementation flexible for future improvements.

V. RESULTS AND DISCUSSIONS

Extensive testing of the proposed intelligent lighting system was conducted in both controlled indoor environments and low-light outdoor conditions to evaluate performance under a variety of scenarios. The primary objective of the testing phase was to assess detection accuracy, angular responsiveness, dimming performance, and overall system stability during continuous operation. The ultrasonic sensors consistently detected obstacles across a practical range of 20 to 250 cm, with an average error ranging between 3 and 5 cm depending on the surface characteristics of the target object. Reflective and curved surfaces occasionally produced minor fluctuations, but the median filtering algorithm minimized the impact of these irregularities, resulting in stable readings across repeated tests.

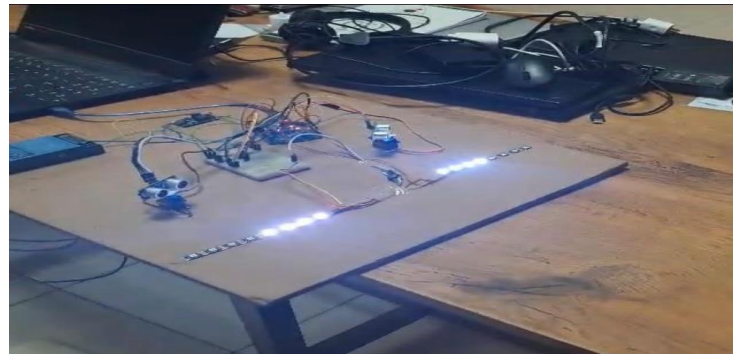


Fig. 4. Semi-LED strip lit-up

The angular scanning mechanism performed reliably, demonstrating smooth servo rotation and accurate mapping of detected angles to

corresponding LED zones. Over multiple full sweeps, the servo maintained positioning accuracy without drift, ensuring that the zonal illumination remained synchronized with the sensed environment. This was particularly important during dynamic testing where objects were repositioned between cycles; the system consistently updated the correct LED segments based on the object's angular location. The overall sweep time for a complete 0° to 180° scan averaged between 100 and 150 ms, allowing the system to refresh environmental data quickly enough to support real-time illumination adjustments.

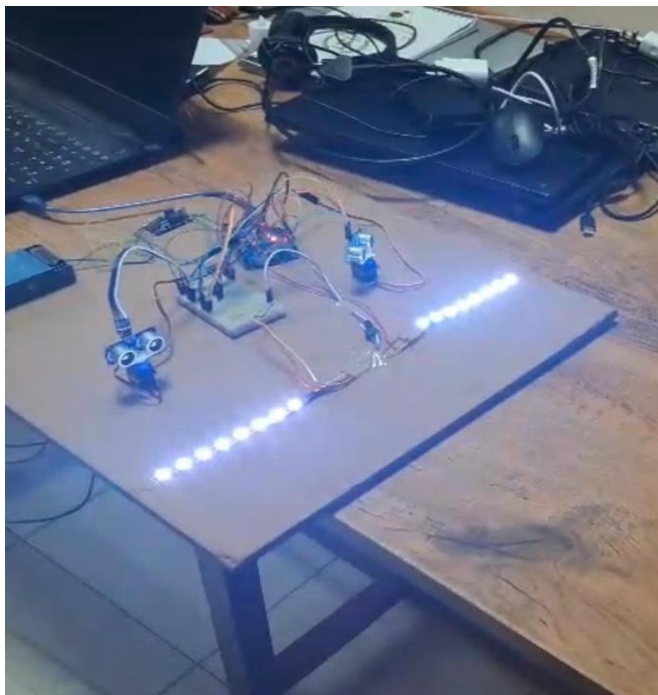


Fig. 5. All LED strip lit-up

Performance of the LED dimming system was evaluated by observing the transitions in brightness across different zones as obstacles were introduced at varying distances and angles. The WS2812 LED strip delivered rapid and smooth brightness adjustments, with response latency consistently measured below 70 ms. This ensured that the system reacted almost instantaneously during scanning, creating an effective adaptive lighting effect. When an obstacle entered the caution or danger zone, only the relevant LED segment dimmed while the remaining zones stayed fully illuminated. This selective dimming behavior closely approximates the functionality of advanced automotive matrix headlights and proved effective in reducing glare directed toward potential hazards.

Outdoor night tests further validated the system's effectiveness. When positioned in an open environment, the LED zones illuminated the simulated road surface clearly, and the dimming mechanism successfully reduced glare when objects were placed in the beam path. Drivers observing the system reported improved clarity in obstacle-free regions and reduced visual strain in areas where dimming occurred. Furthermore, the buzzer provided timely audible feedback whenever objects were detected within the defined danger threshold, adding an additional layer of safety.

Despite the strong performance, some limitations were observed. Ultrasonic sensors were sensitive to soft materials, such as cloth or foam, which occasionally absorbed sound waves and resulted in slightly reduced detection accuracy. Extremely angled surfaces also caused irregular reflections that produced inconsistent distance values in certain cases.

However, these issues did not significantly affect overall system responsiveness, as the filtering mechanism helped stabilize readings. Servo movement speed also imposed a natural limit on how quickly the system could complete a full environmental sweep, suggesting that faster actuation methods or parallel multi-sensor configurations could further improve responsiveness.

Overall, the results confirm that the system achieves its objective of providing adaptive, glare-reducing illumination using low-cost hardware components. The integration of angular scanning, zonal LED control, and real-time decision logic yielded a robust and effective lighting system capable of responding to environmental changes with minimal latency. The performance observed during testing demonstrates the viability of constructing a functional and reliable adaptive lighting solution without relying on high-end sensors or computationally intensive processing techniques.

VI. CONCLUSION AND FUTURE TRENDS

The development of the Intelligent Vehicle Safety System demonstrates that an effective adaptive lighting mechanism can be achieved using low-cost embedded components without relying on high-end automotive sensors. By integrating ultrasonic scanning, servo-based angular positioning, programmable LED zone control, and a synchronized decision-making loop, the system successfully identifies obstacle distance and direction while dynamically adjusting illumination in real time. Testing under indoor and outdoor low-light conditions verified that the system consistently mapped obstacles to the appropriate angular segments, delivering responsive and accurate dimming behavior across the LED zones. The incorporation of caution and danger thresholds ensured that the lighting response was both meaningful and context-aware, while the buzzer alert provided an additional layer of safety during critical proximity conditions. Overall, the results confirm that the proposed design meets its objective of enhancing night-time driving visibility while reducing glare-related hazards.

The overall performance of the system reflects the strength of combining simple sensing techniques with intelligent control strategies. The servo-assisted scanning mechanism enabled a wide field of detection without requiring additional sensors, while the NeoPixel-based LED control ensured smooth and immediate brightness transitions. The project also highlights the practicality of modular embedded design, where each component—sensors, servos, LEDs, power converters, and control logic—contributes to a cohesive and reliable safety solution. Although ultrasonic sensors present limitations when interacting with soft or angled surfaces, the implemented data filtering techniques successfully mitigated most of the noise and inconsistencies encountered during testing. The system remained stable during prolonged operation, indicating robustness suitable for real-world prototyping and further research.

Looking ahead, several advancements can significantly enhance the capabilities of the current system. One major improvement involves integrating camera modules or depth sensors, enabling more detailed environmental understanding through object classification and lane awareness. Combining camera data with ultrasonic scanning could provide hybrid sensing, improving reliability in scenarios where one sensor type encounters limitations. Additionally, implementing machine learning algorithms could allow the system to adapt its illumination pattern based on historical behavioral patterns, road conditions, and object movement predictions. Expanding the LED array into a more granular zone-based matrix would further refine beam shaping, replicating advanced automotive matrix headlights with greater precision.

Furthermore, integrating the system into broader Advanced Driver Assistance Systems (ADAS) could unlock additional functionalities, such as collision prevention, pedestrian detection, and automated

braking support. Communication technologies such as Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) could enable collaborative lighting behavior, where vehicles automatically adjust headlights based on traffic patterns and real-time environmental data. Future developments could also focus on optimizing power consumption and improving environmental durability, making the system compatible with real automotive installation.

In conclusion, the Intelligent Vehicle Safety System establishes a strong foundation for low-cost adaptive lighting research and demonstrates how simple embedded technologies can be combined to mimic high-end automotive features. With enhancements in sensing, intelligence, communication, and scalability, this system has the potential to evolve into a comprehensive smart lighting platform suitable for next-generation intelligent transportation systems.

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